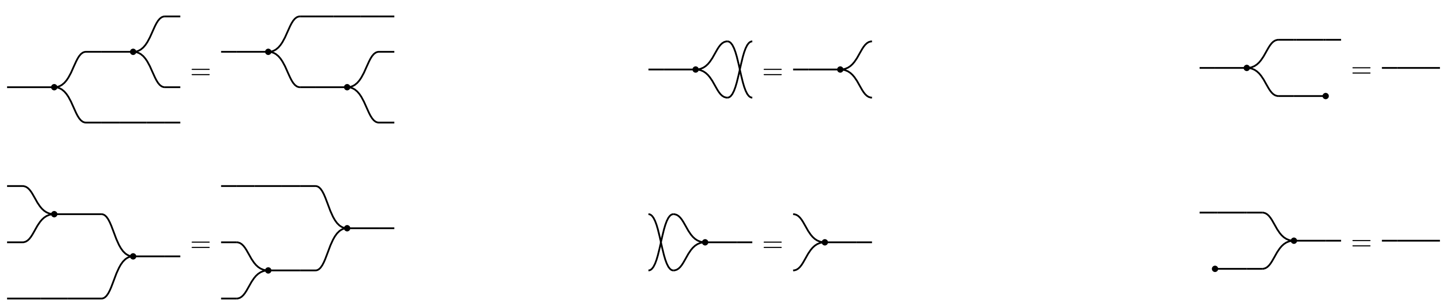
The allegory axioms, and what defines them in the Frobenius calculus

§1. Cartesian bicategory of relations

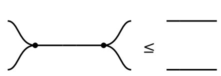
> A **cartesian bicategory of relations** is a poset-enriched category(bicategory) that is has a symmetric monoidal product ⊗ (cartisin product of set in Rel, vertical junctposition in the diagram) and
>
> ∀ object A, `(A, ◁, ⊸) ⊣ (A, ▷, ⟜)`, **Frobenius**, arrows **lax**.
>
> `(◁ : A → A ⊗ A, ⊸ : A → 𝕀)` formed a cocommutative comonoid
>
> `(▷ : A ⊗ A → A, ⟜ : 𝕀 → A)` formed a commutative monoid



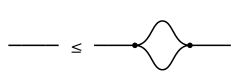
In `Rel(Set)`, the four generators are these relations on a set `A`:

- ◁ `= {(a, (a, a))}` — copy: one in, two identical out
- ▷ `= {((a, a), a)}` — merge: a pair passes only when its two components agree, and the common value comes out
- ⊸ `= A × {*}` — discard: forget the value, keep only that there was one
- ⟜ `= {*} × A` — create: any element at all

§1.1. ∀ object A, `(A, ◁, ⊸) ⊣ (A, ▷, ⟜)`



`▷ ◁ ≤ 𝟙`



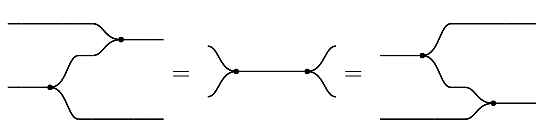
`𝟙 ≤ ◁ ▷`

`⟜ ⊸ ≤ 𝟙`

`𝟙 ≤ ⊸ ⟜`

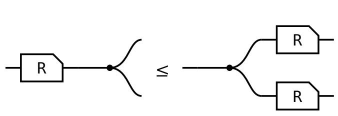
The last of these is the only one that makes a picture **bigger**, and it is worth a name: **a wire is below the cut wire**. In `Rel` it reads `{(a, a)} ⊆ A × A` — cut a wire and its two ends stop having to agree, so cutting can only add pairs. It is the one weakening this calculus gives away for free.

§1.2. The Frobenius law

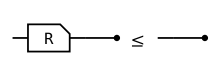


The only clause coupling the comonoid to the monoid beyond adjointness. Both sides reduce to the same picture, ▷ ◁ — merge, then copy — which is the two-in two-out **spider**. Every connected diagram built from these four generators collapses to the spider on its own inputs and outputs, and this equation is what starts that collapse. A **bubble** is the other order, ◁ ▷: copy then merge, a closed loop, which the idempotency of ∩ uses below.

§1.3. Every arrow is a lax comonoid homomorphism



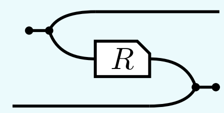
`R◁ ≤ ◁(R⊗R)`



`R⊸ ≤ ⊸`

§2. ° : $\mathcal{C}^{op} \longrightarrow \mathcal{C}$ is a 2 functor

The **converse** R° of R : a → b is R with both of its wires turned round,

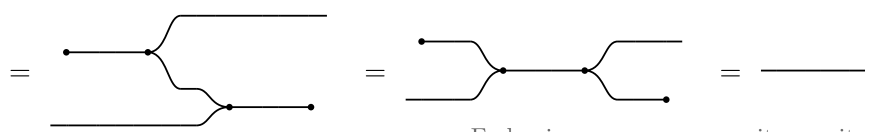


R° = (⊸◁ ⊗ 𝟙) (𝟙 ⊗ R ⊗ 𝟙) (𝟙 ⊗ ▷⊸)

where ⊸◁ : 𝕀 → a ⊗ a opens a pair of wires out of nothing and ▷⊸ : b ⊗ b → 𝕀 closes one, so the input of R° is where the output of R was. And it is a contravariant 2-functor ° : $\mathcal{C}^{op}$ → $\mathcal{C}$:

(i) 𝟙° = 𝟙 (ii) (R S)° = S° R° (iii) (R ⊗ S)° = R° ⊗ S° (iv) R ≤ S implies R° ≤ S°

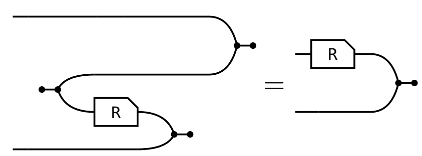
§2.1. 𝟙° = 𝟙 (snake)



Frobenius unit, counit

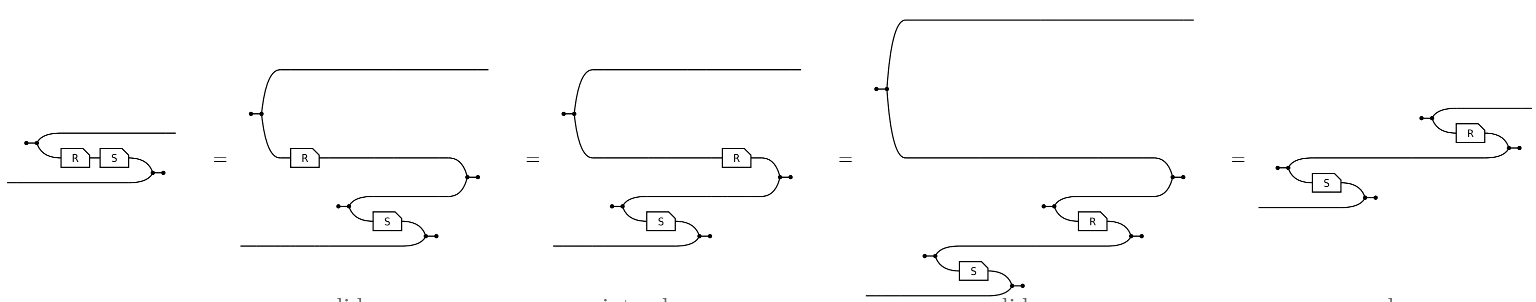
§2.2. The slide

The one rule (ii) and (iii) use, and each of them uses it twice. A converse facing a merge on the lower strand is the box itself, upright, on the upper one:



R° is DEFINED as the bending of (R ⊗ 𝟙) ▷⊸, so the slide claims only that unbending it gives that back — and unbending undoes bending for every arrow. That is the snake above with a passenger: the ⊸◁ bends the a strand down and the ▷⊸ brings it back up, while the b strand rides through untouched. Nothing else is spent below.

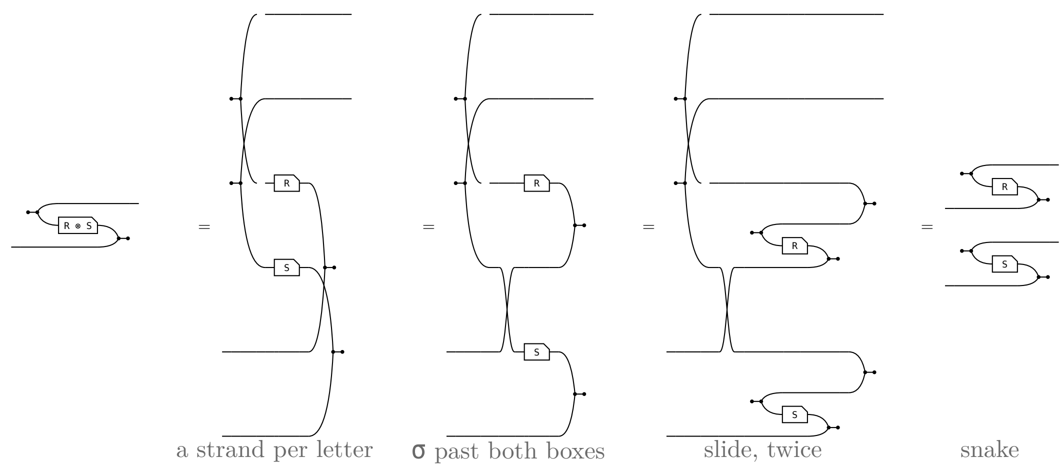
§2.3. (R S)° = S° R°



slide interchange slide snake

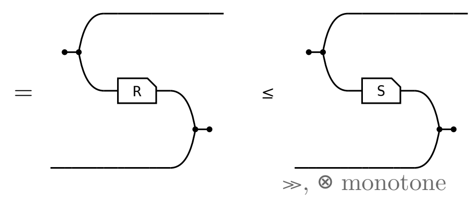
§2.4. (R ⊗ S)° = R° ⊗ S°

A merge at a product is two merges behind a crossing, so the pair unbends one strand at a time and the crossing is all that is left to move.



§2.5. `R ≤ S` implies `R° ≤ S°`

`R` sits in a frame of wires built from ≫ and ⊗, and both of those are monotone, so the box may be replaced where it stands.



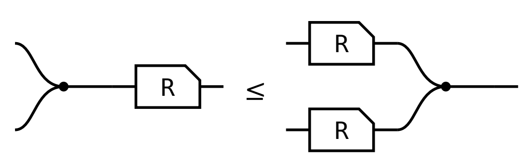
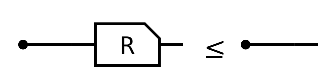
§3. Every arrow is a lax monoid homomorphism

The merge/create mirror of the two comonoid laws opening this note — the paper's (41) and (42), p. 22, which it gets by bending them with (ii) and (iv) above. No bending is needed: expand by the unit `𝟙 ≤ ◁ ▷`, duplicate the box with the comonoid law, contract by the counit `▷ ◁ ≤ 𝟙`. Both halves of the adjunction, and nothing else.

In `Rel(Set)` with `R ⊆ A × B`, where ▷ is the equality test on a pair and ⟜ produces any element at all:

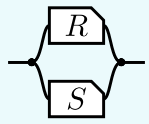
`▷ R`	`= {((a, a ), b) : a R b}` — two equal inputs, then one run of `R`
`(R ⊗ R) ▷`	`= {((a, a'), b) : a R b and a' R b}` — a run on each input, then the results merged
`⟜ R`	`= {(*, b) : ∃a. a R b}` — the image of `R`
`⟜`	`= {*} × B` — all of `B`

Take `a' = a` for the first containment; every `b` lies in `B` for the second. Each is strict exactly when its reverse fails — `R` not injective, `R` not surjective — which is what the later table on maps reads off them.

inequation, and what proves it	picture
(41) `▷ R ≤ (R ⊗ R) ▷` `▷R = ▷R𝟙 ≤ ▷R◁▷ ≤ ▷◁(R⊗R)▷ ≤ (R⊗R)▷` — unit, lax ◁, counit.	
(42) `⟜ R ≤ ⟜` `⟜R = ⟜R𝟙 ≤ ⟜R⊸⟜ ≤ ⟜⊸⟜ ≤ ⟜` — the same three steps at 𝕀.	

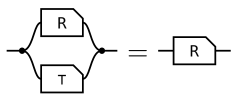
§4. ∩ is a commutative idempotent monoid on every hom-set

The **meet** of $R,\ S : a \to b$, the paper's **convolution**, is $R \cap S := \lhd\ (R \otimes S)\ \rhd$ — copy the input, run R and S on the two copies, merge the results — so what comes out is what both of them do.

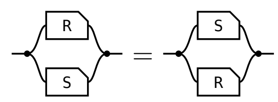


transcribed: a definition has no statement to export

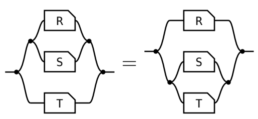
On every hom-set it is associative, commutative and idempotent, with unit the maximal arrow $\top$ = ⊸ ⟜ (the paper's Lemma 4.11).



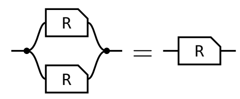
unit: one half of $\top$ per end — the merge's unit law absorbs the ⟜, the copy's counit law the ⊸



commutative: σ crosses $R \otimes S$ by naturality and is absorbed by cocommutativity and commutativity



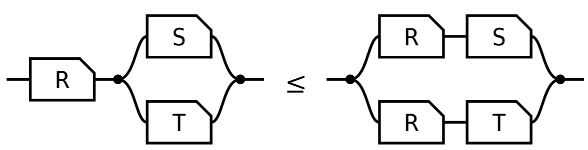
associative: coassociativity and associativity; ⊗ re-brackets for nothing, being strict here



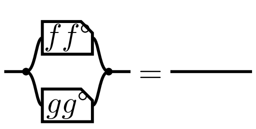
idempotent: the one that is not bookkeeping — the lax copy law is the whole of it, worked in allegory2

So ≤ is the order this monoid induces. $R \cap S \le R$ comes from the unit, and idempotency turns anything under both S and T into something under $S \cap T$, since $R = R \cap R \le S \cap T$.

And one law relating ∩ to composition, which is **not** an equation:

semi-distributivity, and what supplies it	picture
$R\ (S \cap T) \sqsubseteq R\ S \cap R\ T$ — the lax copy law. Equality exactly when R is single valued: the Maps section's $F\ (R \cap S) = F\ R \cap F\ S$.	

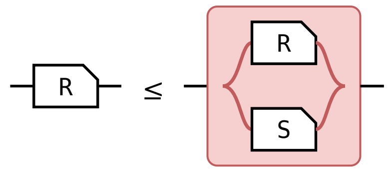
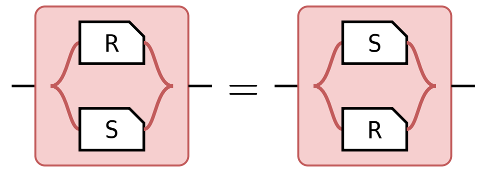
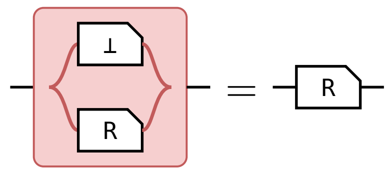
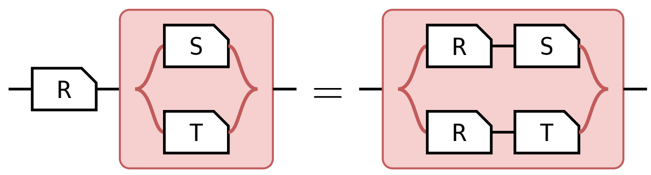
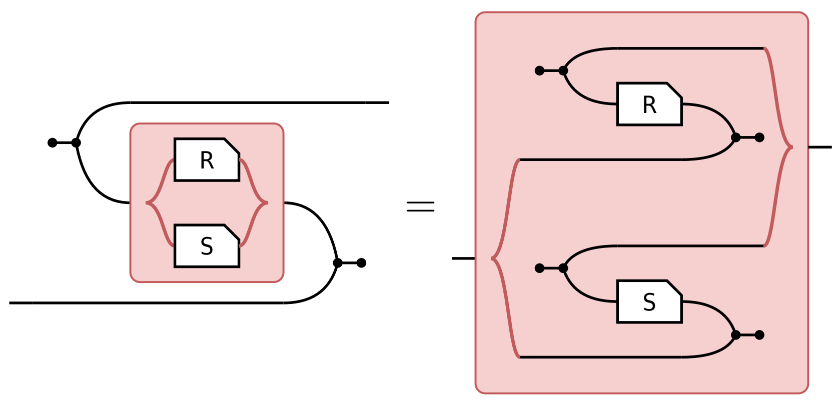
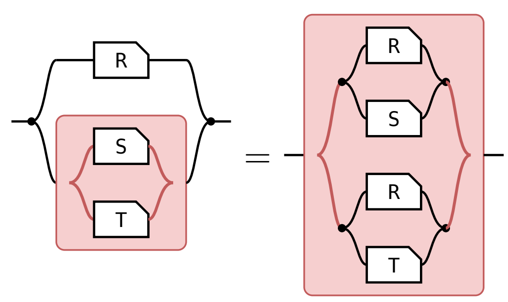
§5. Unitary and pre-tabular — what the other direction needs

condition	allegory	Frobenius side
unitary	$\top$ is a **unit** when $\mathbb{1}\ \top$ is the largest endomorphism on it and every object carries an entire arrow to it. A unit is what makes $\top$ exist.	**free.** $\mathbb{I}$ is the unit object, $\mathbb{1} \le$ ⊸ ⟜ says ⊸ is entire, and $\top$ = ⊸ ⟜ is maximal.
pre-tabular	$f,\ g$ **tabulate** R when both are maps, $R = f°\ g$, and  R is **tabular** when some pair tabulates it, the allegory **pre-tabular** when every arrow lies under a tabular one.	**this is what ⊗ is.** Given unitarity it reduces to tabulating each $\top$, and a tabulation of $\top$ is an object $n \otimes m$ with its two projections. One side derives the product from tabulations, the other posits it.

Together: **cartesian bicategory of relations ≃ unitary pre-tabular allegory**.

§6. Union

Past what the definition can build: a union needs a **biproduct** on top of the Frobenius structure. It draws as a **tape** — the rounded wrapper is the second product, and its fork and join open and close a branch a particle takes exactly one of. The residual, which needs a second composition with linear adjoints, is in the division section below.

statement	picture
`R ⊑ R ∪ S` `union R S := ⟨R, S⟩ [𝟙, 𝟙]`: offer both branches, then forget which was taken.	
`R ∪ S = S ∪ R`	
`⊥ ∪ R = R` ⊥ routes through the zero object; there is no shape for it.	
`R (S ∪ T) = R S ∪ R T` — composition distributes over union, which `∩` does not.	
`(R ∪ S)° = R° ∪ S°`	
`R ∩ (S ∪ T) = (R ∩ S) ∪ (R ∩ T)` — what makes the whole **distributive**, and the one picture with both layers in it.	

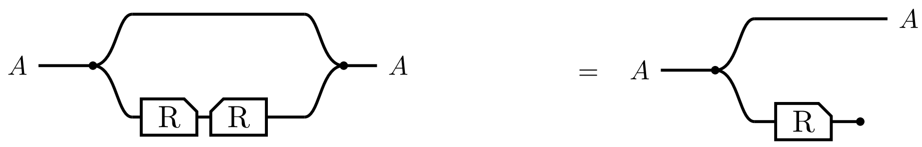
The last of Freyd's §2.21 equations that have no picture here — a tape around nothing, or a tape around what is already inside it, is not worth drawing. `R` = shops with rice, `S` = shops with sugar throughout.

equation	the rice reading
`R ∪ (S ∩ R) = R` `(R ∪ S) ∩ R = R` absorption	Shops with rice, or with both: still the shops with rice. Shops with rice or sugar, and with rice: the same. The smaller side of each pair is already inside the larger.

equation	the rice reading
`R ⊥ = ⊥` `R 0_S = 0_{R S}` — and `⊥ R = ⊥` on the other side	Going to a shop and then taking a road that leads nowhere gets you nowhere. `⊥` is a two-sided zero for composition, as `⊤` is not.

§7. Domain and range

> The **domain** `Dom R ≜ 𝟙 ∩ R R°` and the **range** `Ran R ≜ Dom R°`.



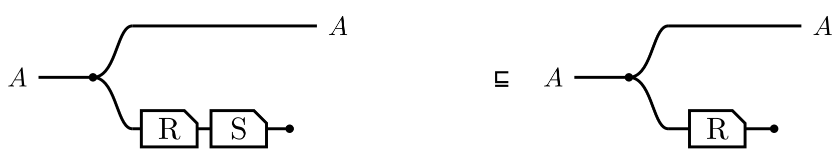
▷ lands the return leg back on the value ◁ handed out,
so `R°` ▷ cuts to ⊸ — Frobenius

Running `R` and throwing the result away leaves only the fact that `R` could fire, and `Ran R` is the same picture with the box mirrored. In `Rel` both steps are `{(a,a) : ∃b. a R b}`.

`Dom R ≜ 𝟙 ∩ R R°`
`Dom R ⊑ A` ⟺ `R ⊑ A R`, for `A` coreflexive
`Dom (R S) ⊑ Dom R`
`Dom (R ∩ S) = 𝟙 ∩ S R°`
`R` entire ⟺ `Dom R = 𝟙` ⟺ `𝟙 ⊑ R R°`
`R` simple ⟺ `R° R ⊑ 𝟙`
`R` a map ⟺ `R` entire and simple
`R, S` entire ⟹ `R S` entire — likewise simple, likewise maps
`R S` entire ⟹ `R` entire

§7.1. Sliding the discard

`Dom (R S) ⊑ Dom R`, and a single glyph for `Dom` would have nothing to slide: with the box and the discard drawn apart, the law is one dot walking back along the lower strand.

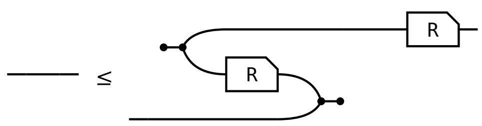


`S` ⊸ ⊑ ⊸, the lax axiom for ⊸ in the first section
— the discard slides back past `S`

Equality is `S` entire, which is the same picture read as `Dom R = 𝟙` ⟺ `R` entire.

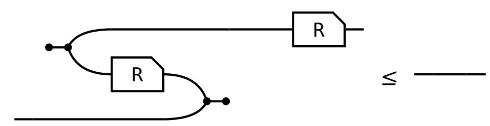
§8. Maps

Every arrow is lax for ◁ and for ⊸ by axiom, and for ▷ and ⟜ by the lax monoid section. All four hold for **every** arrow, their REVERSES do not, and each reverse holding is a property of the arrow. The right-hand column is the **adjoint** form, which is the definition used here because it is literally the allegory's `Simple` and `Entire`; that the two forms agree is a separate theorem, not proved here.



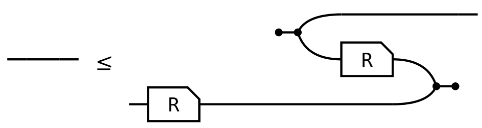
surjective

$\mathbb{1} \sqsubseteq R^\circ\ R$, that is, R° entire.



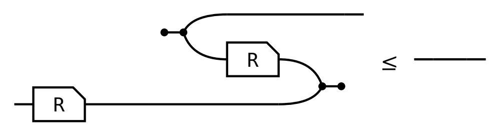
single valued

$R^\circ\ R \sqsubseteq \mathbb{1}$



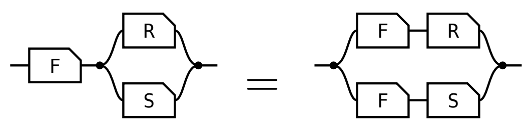
entire

$\mathbb{1} \sqsubseteq R\ R^\circ$. With **single valued**, a **map**.



injective

$R\ R^\circ \sqsubseteq \mathbb{1}$

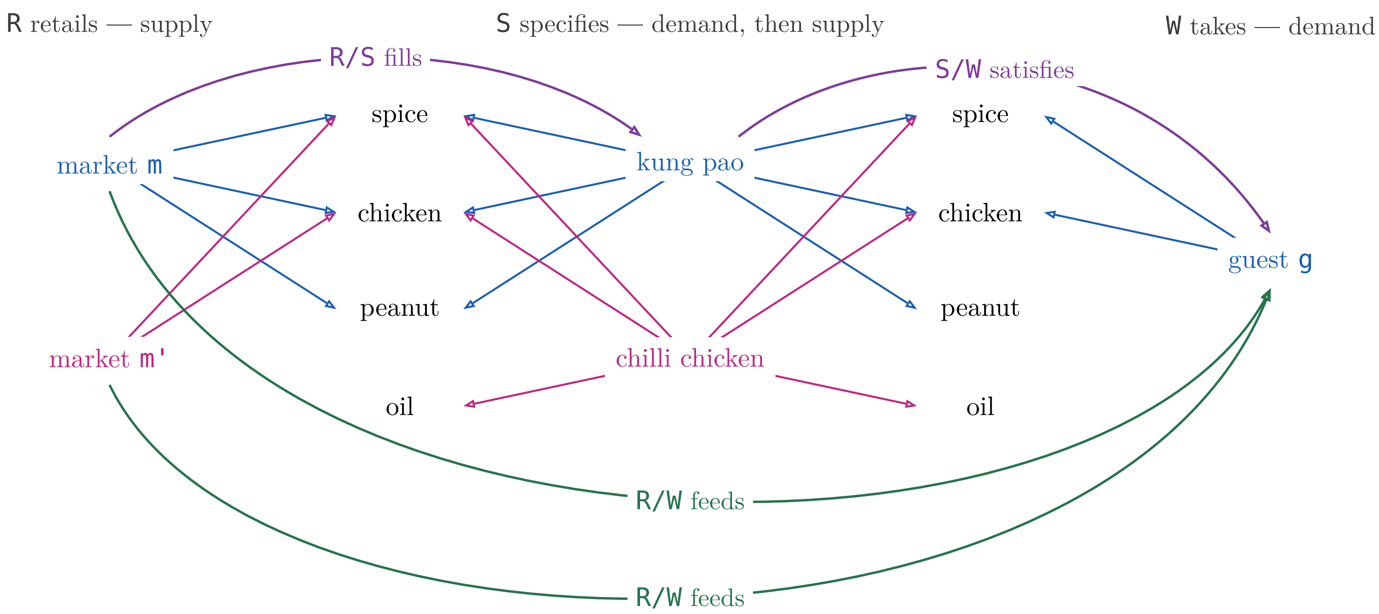
law	picture
`F (R ∩ S) = F R ∩ F S` F single valued F = shops Ann may go to, R = has rice, S = has sugar. Single valued means one shop, so the sides agree.	 one shop with both — rice at A, sugar at B

§9. Division is WHAT!

> `m (R/S) d ⟺ ∀i. d S i → m R i`
>
> `m's image contains d's pre image.`

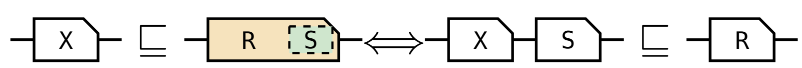
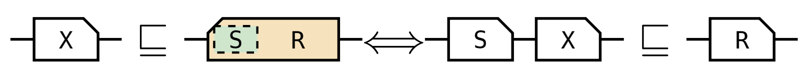
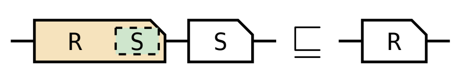
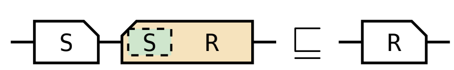
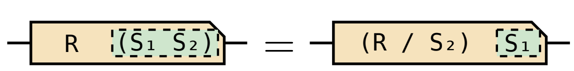
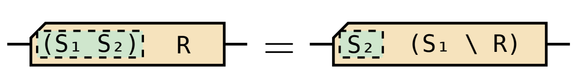
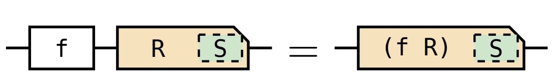
`R/S` relates a producer to a consumer when the producer supplies *WHAT* the consumer asks for. m' fills no dish, even it has everything the guest wants.

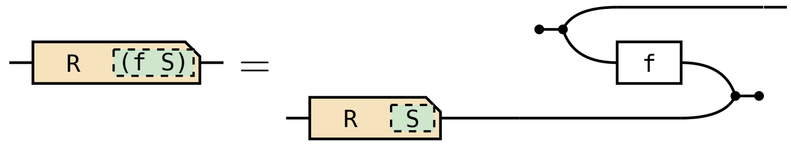
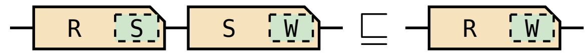
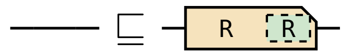
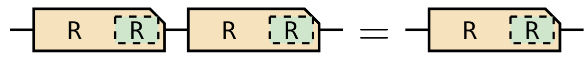
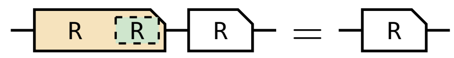
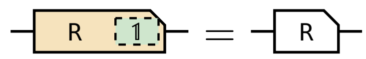
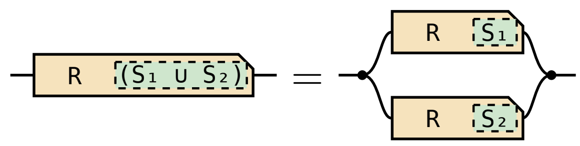
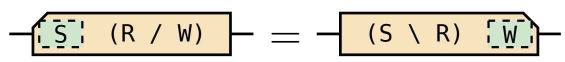
§9.1. `(R/S)(S/W) ⊑ R/W`



`R/S` — market retails *what* dish specifies   `S/W` — dish specifies *what* guest takes
`R/W` — market retails *what* guest takes

`(R/S)(S/W)` is a path: `m` → kung pao → `g`, and that is all of it. `R/W` also holds of `m'`, which retails everything `g` wants — but `m'` fills no dish, so nothing composes to it. The missing path is exactly the strictness of `(R/S)(S/W) ⊑ R/W`.

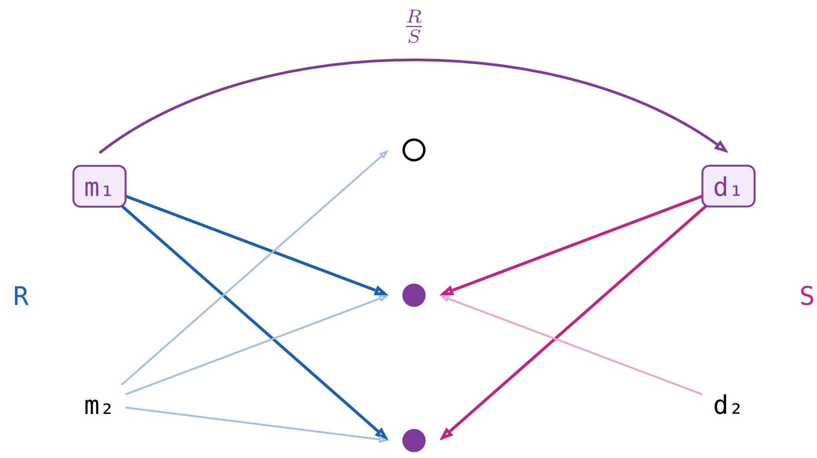
`X ⊑ R/S ⟺ X S ⊑ R` `X` is any market-to-dish pairing; one that only pairs a market with dishes it fills lies inside `R/S`, and `R/S` is the largest such.	
`X ⊑ S\R ⟺ S X ⊑ R` The mirror — divide on the left when the market comes first.	
`(R/S) S ⊑ R` There is a dish `m` fills that specifies `i` — then `m` retails `i` too. Strict at `S = ∅`: `R/S` is everything, `(R/S) S = ∅`.	
`S (S\R) ⊑ R` The mirror.	
associate: $R/(S_1\ S_2) = (R/S_2)/S_1$ A dish of dishes: divide by the far end first.	
$(S_1\ S_2)\backslash R = S_2\backslash(S_1\backslash R)$ The mirror.	
maps: `f (R/S) = (f R)/S` Rename the market before or after dividing — the licence to write `f R / S`.	

$R/(f\ S) = (R/S)\ f°$ Rename the dish: a map leaves a denominator as $f°$ outside the box.	
$(R/S)(S/W) \sqsubseteq R/W$ A market that fills a dish that satisfies a guest feeds that guest.	
$\mathbb{1} \sqsubseteq R/R$ R/R runs market to market: each fills its own list. Strict: two markets retailing only spice fill each other's and stay two markets.	
$(R/R)(R/R) = R/R$ R/R is the preorder **retails at least as much as**, and a preorder is idempotent. Freyd writes $\sqsubseteq$; with $\mathbb{1} \sqsubseteq R/R$ above it is an equality.	
$(R/R)\ R = R$ Reaching i through a market whose list m fills is reaching it directly, since m fills its own.	
$R/\mathbb{1} = R$ Filling the dish that specifies exactly i is retailing i.	
$R/(S_1 \cup S_2) = R/S_1 \cap R/S_2$ Filling a dish made of two is filling each of them.	
$S\backslash(R/W) = (S\backslash R)/W$ Which is why $S\backslash R/W$ needs no bracket.	

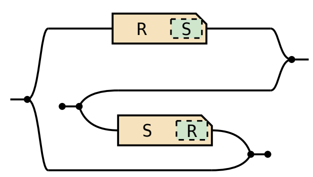
Fifteen laws, fifteen pictures, and not one shows a generator: ∩, ∪, ° and composition are what the Frobenius generators build, and / is none of those — it is posited, with nothing to unfold.

§10. Symmetric division

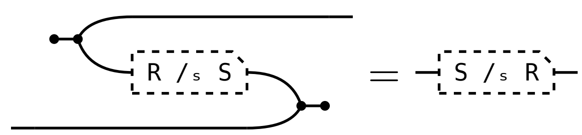
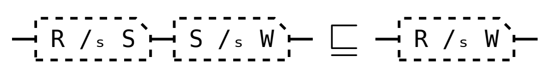
> $\frac{R}{S}$ ≜ `(R/S) ∩ (S/R)°`. In `Rel` `m` and `d` has the same image: `∀i. (m R i ⟺ d S i)`



A meet of two long divisions, the second turned round by the converse frame:



exported from the definition, not transcribed

$\left(\frac{R}{S}\right)^{\circ} = \frac{S}{R}$ Matching is symmetric.	
$\frac{R}{S}\frac{S}{W} \sqsubseteq \frac{R}{W}$ And transitive.	

No pictures for the rest of §2.35: symmetric division is not built from the generators.

law	**the market reading**
$X \sqsubseteq \frac{R}{S} \iff XS \sqsubseteq R$ and `X° R ⊑ S`	`X` may pair `m` with `d` only when each fills the other. Both halves must typecheck, so the operation is **partial**.
$\frac{R}{S}S \sqsubseteq R$	$\frac{R}{S}S = \mathtt{Dom}(\frac{R}{S})R$, which is `R` with the unmatched markets cut out.
$\frac{R}{R}R \sqsubseteq R$	The same with `R` against itself, market to market.
$\mathbb{1} \sqsubseteq \frac{R}{R}$	m_1 R {i1,i2} R° m_2 , 2 different market can have the same image and create extra pair than 𝟙
$\left(\frac{R}{R}\right)^2 = \frac{R}{R}$	So **matches** is an equivalence relation. Freyd writes `⊑`; with the row above it is an equality.
$X \sqsubseteq \frac{R}{R} \iff XR \sqsubseteq R$, for symmetric `X`	The largest symmetric arrow that leaves `R` alone.
$\frac{R}{\mathbb{1}}$ is the **simple part** of `R`	The markets retailing one ingredient and nothing else. It equals `R` only when `R` is simple, unlike `R/𝟙 = R`.
`Dom` $\frac{R}{S}$ `= 𝟙 ∩ (R/S)(S/R)`	Its domain is the **domain of simplicity** of `R`.

§10.1. Straight

> `S` is **straight** when $\frac{S}{S} = \mathbb{1}$ — no two dishes specify the same ingredients.

law	the market reading
every symmetric `X` with `X S ⊑ S` is coreflexive	Equivalently, `S` is straight.
`f S = g S` ⟹ `f = g`	A straight `S` tells its dishes apart, so it cancels on the right.
`S R` straight ⟹ `S` straight	If the longer chain tells them apart, the first step already does.
`S` straight ⟺ `R/S` simple for all `R`	If no two dishes agree, at most one market can match a given dish.
`R = h S`, `h` a cover, `S` straight	In an effective division allegory every arrow factors that way.

§11. Power allegories

A **power allegory** is a division allegory with one unary operation on arrows, ∋ **epsiloff**, subject to

$$\ni_R\Box = R\Box, \quad \ni_R = \ni_{R\,\Box}$$
$$\mathbb{1} \sqsubseteq (R/\ni_R)(\ni_R/R) \qquad \ni_R \text{ is } \textbf{thick}$$
$$(\ni_R/\ni_R) \cap (\ni_R/\ni_R)^\circ \sqsubseteq \mathbb{1} \qquad \ni_R \text{ is } \textbf{straight}$$

`R □` is R's target, an identity arrow. For `R : A → B` write `∋ : [B] → B`, dropping the subscript.

In `Rel` its source is the powerset of its target and `l ∋ i` iff `i ∈ l`, and `∋` reads a list `l` back into the ingredients on it.

law	the market reading
$\ni_R\Box = R\Box, \quad \ni_R = \ni_{R\,\Box}$	`∋` has the same target as `R`, and replacing `R` by the identity at that target leaves it unchanged: one `∋` per object, not per arrow.
`∋` is **thick**	**Comprehension**: every market has a list of exactly what it retails. Equivalently every `R` factors as a map followed by `∋`.
`∋` is **straight**, that is $\frac{\ni}{\ni} \sqsubseteq \mathbb{1}$	**Extensionality**: two lists with the same ingredients are the same list.
`Λ(R)` ≜ $\frac{R}{\ni}$ `: A → [B]`, for `R : A → B`	The list-of map: send a market to its list.
`Λ(R)` is simple	`∋` is straight, and dividing by a straight arrow is simple. At most one list per market.
`Λ(R)` is entire ⟺ `∋` thick	`Dom` $\frac{R}{\ni}$ `= 𝟙 ∩ (R/∋)(∋/R)`, the domain row above. At least one list per market, so `Λ(R)` is a **map**.
`Λ(R) ∋ = R`	Look up a market's list, then read off its ingredients: what it retails.
`Λ(R)` is the only map with `Λ(R) ∋ = R`	Two maps naming the same ingredients name the same list — extensionality again.
`F ⊑ Λ(F ∋)`, `F` simple	A partial choice of lists is inside the total one.
`[α]` = source of `∋`, the **power-object**	Every list over `α`.
`{·}` ≜ `Λ(𝟙)`, the **singleton map**, monic	The one-ingredient list. `Λ(𝟙)Λ°(𝟙) ⊑` $\frac{\mathbb{1}}{\ni}\frac{\ni}{\mathbb{1}} \sqsubseteq \frac{\mathbb{1}}{\mathbb{1}} \sqsubseteq \mathbb{1}$.
`Λ(f) = f {·}`, `f` a map	Rename first or take singletons first.
`C` a topos ⟹ `Rel(C)` a power allegory	And back: a unitary tabular power allegory has `Map(A)` a topos. Extensionality and comprehension are all a topos adds.

§12. Relator

> Every hom-set of an allegory is a poset, so an allegory is a **locally posetal 2-category**: the 2-cell from R to S IS $R \sqsubseteq S$. A **relator** $F : \mathcal{C} \to \mathcal{D}$ is a 2-functor between allegories:
>
> $$F\ \mathbb{1} = \mathbb{1} \qquad F(R\ S) = (F\ R)(F\ S) \qquad R \sqsubseteq S \implies F\ R \sqsubseteq F\ S$$
>
> Preserving $°$ is **not** asked for — $°$ is an identity-on-objects involution $\mathcal{C}^{op} \to \mathcal{C}$, no part of the 2-category.

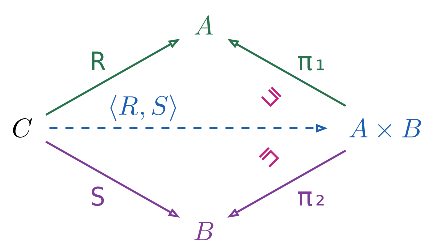
- For f a map, $F\ f$ is a map and $F(f°) = (F\ f)°$.
- Over a **tabular** allegory a functor is a relator $\iff$ it preserves $°$.
- A relator is fixed by what it does to maps.
- $F(R \cap S) \sqsubseteq (F\ R) \cap (F\ S)$, and strictly.
- $F(X \cap Y) = (F\ X) \cap (F\ Y)$ for $X,\ Y$ coreflexive.
- $F(\mathrm{Dom}\ R) = \mathrm{Dom}(F\ R)$ for F preserving $°$.

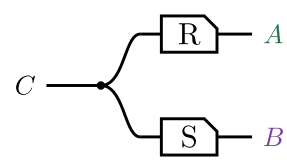
The **power relator** P — $x\ (P\ R)\ y \iff (\forall a \in x.\ \exists b \in y.\ a\ R\ b) \wedge (\forall b \in y.\ \exists a \in x.\ a\ R\ b)$ — is where the fourth is strict: for $R = \{(a_1,b_1),\ (a_2,b_2)\}$ and $S = \{(a_1,b_2),\ (a_2,b_1)\}$ the pair $(\{a_1,a_2\},\ \{b_1,b_2\})$ is in $P\ R \cap P\ S$, while $R \cap S = \varnothing$.

§12.1. The pairing $\langle R,S\rangle$

> The **pairing** of $R : C \to A$ and $S : C \to B$ is $\langle R,S\rangle \triangleq R\ \pi_1° \cap S\ \pi_2°$, where $(\pi_1,\ \pi_2)$ is the tabulation of $\top$.

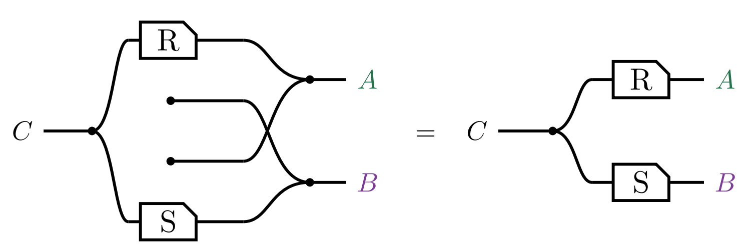
$$\langle R,S\rangle\ \pi_1 = (\mathrm{Dom}\ S)\ R \qquad \langle R,S\rangle\ \pi_2 = (\mathrm{Dom}\ R)\ S$$





A domain is coreflexive, so $\langle R,S\rangle\ \pi_1 \sqsubseteq R$, with equality exactly when S is entire; for maps both triangles commute and $\langle f,g\rangle$ is unique. In Rel, $c\ \langle R,S\rangle\ (a,b)$ iff $c\ R\ a$ and $c\ S\ b$ — copy c, then R on one strand and S on the other, which is $\lhd\ (R \otimes S)$ on the right.

No $°$ survives the translation. $\pi_1 = \mathbb{1} \otimes \multimap$ discards the second component, so $\pi_1° = \mathbb{1} \otimes \multimapinv$ **creates** one out of nothing, and $\cap$ is copy, run both, merge. Draw that and the created strands — the two dots with no left end, and the crossing they force — are merged against real ones, which is the monoid's unit law:

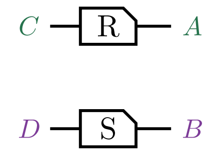


$\multimapinv\ \rhd = \mathbb{1}$ on each half

§12.2. The relational product $R \times S$

> $R \times S \triangleq \langle \pi_1\ R,\ \pi_2\ S\rangle$, a relator in each argument but no longer a categorical product.

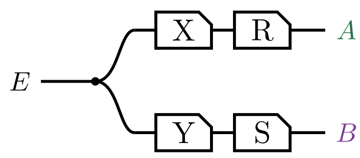
$$\begin{tikzcd}
C \arrow[rr, "R"] & & A \\
C \times D \arrow[u, "\pi_1"] \arrow[d, "\pi_2"'] \arrow[rr, dashed, "R \times S"] & & A \times B \arrow[u, "\pi_1"'] \arrow[d, "\pi_2"] \arrow[u, phantom, "\sqsubseteq", shift right=6] \arrow[d, phantom, "\sqsubseteq", shift right=6] \\
D \arrow[rr, "S"'] & & B
\end{tikzcd}$$



Right-then-up is $(R \times S)\ \pi_1$, up-then-right is $\pi_1\ R$, and $(R \times S)\ \pi_1 \sqsubseteq \pi_1\ R$, equality when S is entire. In Rel, $(c,d)\ (R \times S)\ (a,b)$ iff $c\ R\ a$ and $d\ S\ b$ — two strands side by side, no copy dot: $R \times S = R \otimes S$.

§12.3. Absorption

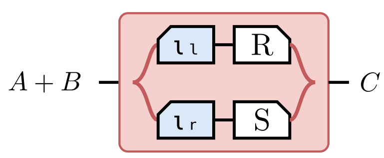
For $X : E \to C$ and $Y : E \to D$, $\langle X,Y\rangle\ (R \times S) = \langle X\ R,\ Y\ S\rangle$. Both sides are this picture:



§12.4. The coproduct [R,S]

The injections $\iota_l : A \to A + B$ and $\iota_r : B \to A + B$ are maps, and the coproduct they make of the maps stays a coproduct once every arrow is allowed: both equations hold with equality and $[R,S]$ is the only arrow satisfying them, with none of the Dom slack $\langle R,S\rangle$ carries.

> $[R,S] \triangleq \iota_l^\circ\ R \cup \iota_r^\circ\ S$, and $R + S \triangleq [R\ \iota_l,\ S\ \iota_r]$.



The tape is the union — a particle entering at $A + B$ takes exactly one branch — and the two mirrored boxes are what makes the branches disjoint.

$\iota_l\ [R,S] = R$, $\iota_r\ [R,S] = S$, and $[R,S]$ is the only such arrow
$\iota_l\ \iota_l^\circ = 1\!\!1 = \iota_r\ \iota_r^\circ$
$\iota_l\ \iota_r^\circ = \bot = \iota_r\ \iota_l^\circ$
$\iota_l^\circ\ \iota_l \cup \iota_r^\circ\ \iota_r = 1\!\!1$
$[U,V]^\circ\ [R,S] = U^\circ\ R \cup V^\circ\ S$

The first row is not free: $\iota_l,\ \iota_r$ were only ever asked to be a coproduct of **maps**. They stay one once every arrow is allowed because Λ sends an arrow $A \to C$ to a map $A \to [C]$ reversibly, so the map coproduct can be applied underneath it. For any $T : A + B \to C$,

$$\begin{aligned}
& \iota_l\ T = R \text{ and } \iota_r\ T = S \\
\Longleftrightarrow\quad & \text{Λ is injective — $\Lambda(X)\ \ni\ = X$ gives X back} \\
& \Lambda(\iota_l\ T) = \Lambda(R) \text{ and } \Lambda(\iota_r\ T) = \Lambda(S) \\
\Longleftrightarrow\quad & \text{Λ fusion, $\Lambda(f\ X) = f\ \Lambda(X)$ for f a map} \\
& \iota_l\ \Lambda(T) = \Lambda(R) \text{ and } \iota_r\ \Lambda(T) = \Lambda(S) \\
\Longleftrightarrow\quad & \text{the coproduct of maps, at the map $\Lambda(T)$} \\
& \Lambda(T) = [\Lambda(R),\ \Lambda(S)] \\
\Longleftrightarrow\quad & \text{$\Lambda(X)$ is the only map with $\Lambda(X)\ \ni\ = X$} \\
& T = [\Lambda(R),\ \Lambda(S)]\ \ni
\end{aligned}$$

Every step is an $\Longleftrightarrow$, so $[\Lambda(R),\ \Lambda(S)]\ \ni$ is the **only** arrow satisfying the two equations — which is what $[R,S]$ was claimed to be. Drawn:

$$\begin{tikzcd}
 & A \arrow[dl, "\iota_l"'] \arrow[dr, "\Lambda(R)"] \arrow[rr, bend left, "R"] & & \\
A+B \arrow[rr, dashed, "{[\Lambda(R),\ \Lambda(S)]}"] & & {[C]} \arrow[r, "\ni"] & C \\
 & B \arrow[ul, "\iota_r"] \arrow[ur, "\Lambda(S)"'] \arrow[urr, bend right, "S"'] & &
\end{tikzcd}$$

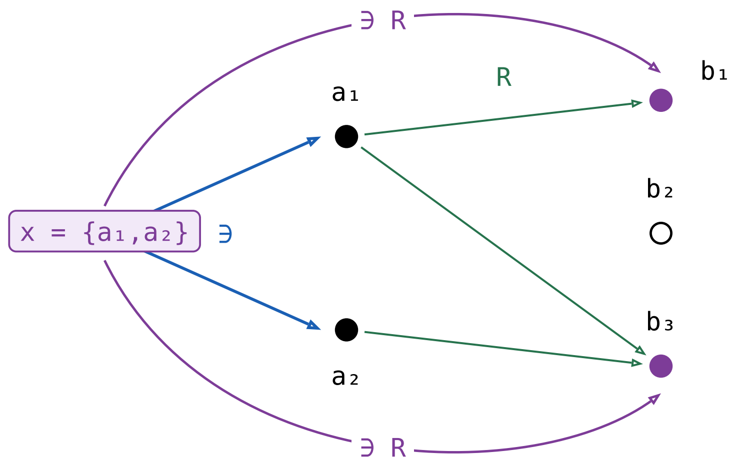
Nothing here holds only up to $\sqsubseteq$: every triangle commutes on the nose, which is the difference from the pairing above. The border spells $[R,S] = [\Lambda(R),\ \Lambda(S)]\ \ni$, and pushing $\ni$ into the union that $[\cdot,\cdot]$ on maps already is turns that back into the definition, $(\iota_l^\circ\ \Lambda(R) \cup \iota_r^\circ\ \Lambda(S))\ \ni\ = \iota_l^\circ\ R \cup \iota_r^\circ\ S$.

§12.5. The power relator `P R`

For `R : A → B`, `P R ≜ ((∋ R)/∋) ∩ ((∋ R°)/∋)° : [A] → [B]`.

`∋ R` is a **composition** — `∋ : [A] → A` followed by `R` — and not Freyd's subscripted $\ni_R$, which is the `∋` at `R`'s **target**: §11 writes one `∋` per object and drops the subscript. Piece by piece, on one instance, with `x, y` lists and `a, b` their elements.

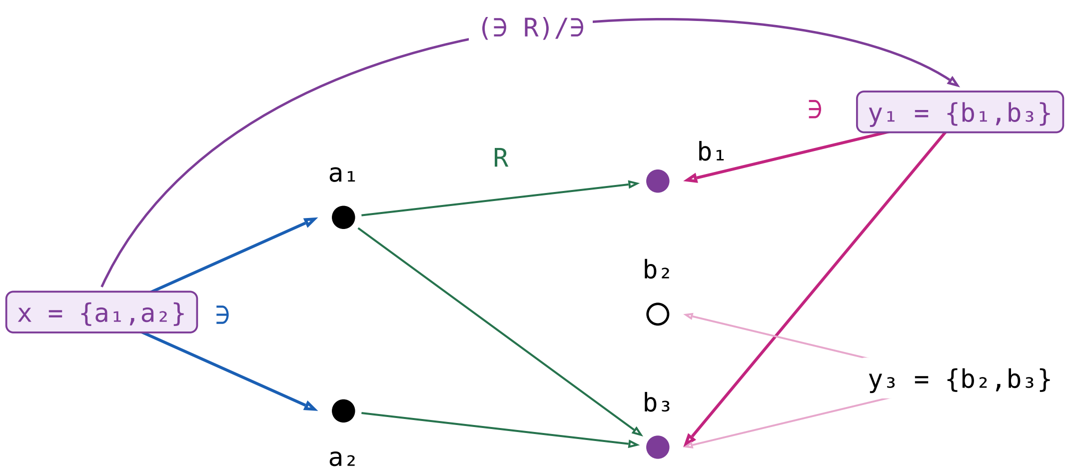
`x = {a₁,a₂}`, and `R` sends `a₁` to `b₁` and `b₃`, `a₂` to `b₃` alone. Nothing on `x` reaches `b₂`, so `b₂` is drawn hollow.



`∋ R : [A] → B`   `x (∋ R) b ⟺ ∃a ∈ x. a R b`

one arc per element of `B` that `x` reaches, and `b₂` gets none

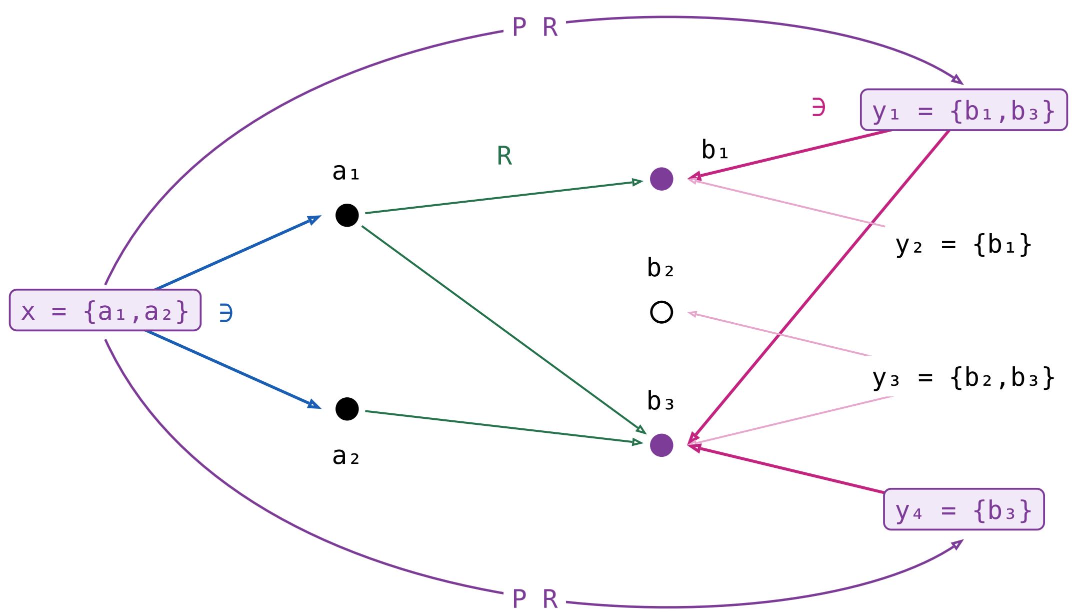
Dividing by `∋` turns that into a relation between **lists**: every element of `y` must be one of the filled dots.



`(∋ R)/∋ : [A] → [B]`   `x ((∋ R)/∋) y ⟺ ∀b ∈ y. ∃a ∈ x. a R b`

`y₃` is rejected: it names `b₂`, which `x` does not reach

The other half of the meet is that clause with the lists swapped and R turned round: every element of x must reach y. Both together, and two more lists to separate them:



$$((\ni\ R^\circ)/\ni)^\circ : [A] \to [B] \qquad x\ ((\ni\ R^\circ)/\ni)^\circ\ y \iff \forall a \in x.\ \exists b \in y.\ a\ R\ b$$

y_2 is rejected too: a_2's only image is b_3, which y_2 does not name

In words, if $x\ (P\ R)\ y$ then every element of x is related by R to some element of y, and conversely — the two clauses of $\cap$, one per direction.

Neither $\ni$ nor $P\ R$ is a map. $\ni$ sends a list to **each** of its elements; $P\ R$ sends x to **every** y meeting the two clauses, y_1 and y_4 both. Nor is it entire: an $a \in x$ with no image at all leaves x with no partner whatever. Turning a relation into a map is Λ's job (§11), and on lists that map is $\Lambda(\ni\ R)$ — of the four it accepts y_1 only.

Not a symmetric division. $\frac{\ni R}{\ni}$ is $\Lambda(\ni\ R)$, and in $\mathtt{Rel}$ it reads $x\ \Lambda(\ni\ R)\ y \iff y = \{b \mid \exists a \in x.\ a\ R\ b\}$: in words, y is **exactly** what x reaches, where $P\ R$ asks only that each side cover the other. Nor can it be repaired: in that fraction's second half $(\ni/(\ni R))^\circ$ the R sits in a denominator, so the operation is antitone there, and a relator has to be monotone. $P\ R$ keeps the first half and takes the second half at R°, which puts R back in a numerator. The fraction returns exactly where the two halves agree — at $\mathbb{1}$, and at a map.

law	the reading
$X \sqsubseteq P\ R \iff X\ \ni\ \sqsubseteq\ \ni\ R$ and $X^\circ\ \ni\ \sqsubseteq\ \ni\ R^\circ$	One containment, and the same one at R° — which is the definition read off the two divisions. Hence $P(R^\circ) = (P\ R)^\circ$, and $R \sqsubseteq S \implies P\ R \sqsubseteq P\ S$.
$P\ \mathbb{1} = \frac{\ni}{\ni} = \mathbb{1}$	The straightness axiom verbatim: extensionality **is** P's unit law.
$P\ f = \Lambda(\ni\ f) = \frac{\ni f}{\ni}$, for f a map	In $\mathtt{Rel}$, $x\ (P\ f)\ y \iff y = \{f\ a \mid a \in x\}$. The half at f° says every $a \in x$ has its $f\ a$ on y; f has just the one image per a, so that already says y contains everything x reaches, which is the fraction's second half. For a map the two definitions coincide.
$P(R\ S) = (P\ R)(P\ S)$	$\sqsupseteq$ is the division cancellation laws. $\sqsubseteq$ is the one law in this section that is not a calculation: it needs a tabulation of $P(R\ S)$.

§13. Catamorphism

> F a relator with an **initial algebra** $\alpha : F\ T \to T$ among the maps. For a relational algebra $R : F\ B \to B$, the **catamorphism** $\llparenthesis R \rrparenthesis : T \to B$ is the unique arrow with $\alpha\ \llparenthesis R \rrparenthesis = (F\ \llparenthesis R \rrparenthesis)\ R$.

$$\begin{tikzcd}
F\ T \arrow[r, dashed, "F\ X"] \arrow[d, "\alpha"'] & F\ B \arrow[d, "R"] \\
T \arrow[r, dashed, "X"'] & B
\end{tikzcd}$$

With $X = \llparenthesis R \rrparenthesis$ the square commutes strictly: unlike the product's triangles, there is no slack here, so no $\sqsubseteq$ appears anywhere in it.

The three **fusion** rows rewrite $\llparenthesis R \rrparenthesis\ S$ through a second algebra $Q : F\ C \to C$ along an arrow $S : B \to C$. They are the only rows here that need the allegory **locally complete** — every hom-set a complete lattice — because they come from a least-fixed-point argument; the rest of the table needs only the initial algebra.

name	**law**	**what it says**
the defining equation	$X = \llparenthesis R \rrparenthesis \iff \alpha\ X = (F\ X)\ R$	Folding a value that α has just built is the same as folding its parts and combining them with R. Nothing else has that property.
Lambek	$\alpha^\circ = \alpha^{-1}$, the initial algebra is an isomorphism	A constructor can be undone — α° takes a value apart into the parts it was built from.
Eilenberg–Wright	$\Lambda\llparenthesis R \rrparenthesis = \llparenthesis \Lambda((F\ \ni)\ R) \rrparenthesis$	A relational fold is a deterministic fold of SETS: Λ pushes the nondeterminism into the power-object, where the fold is a map again.
Eilenberg–Wright	$\llparenthesis R \rrparenthesis = \llparenthesis \Lambda((F\ \ni)\ R) \rrparenthesis\ \ni$	The same fact read back — fold deterministically into a set, then take a member of it.
fusion	$\llparenthesis Q \rrparenthesis \sqsubseteq \llparenthesis R \rrparenthesis\ S \Longleftarrow (F\ S)\ Q \sqsubseteq R\ S$	Half of fusion: an inclusion between the two algebras is inherited by the folds.
fusion	$\llparenthesis R \rrparenthesis\ S \sqsubseteq \llparenthesis Q \rrparenthesis \Longleftarrow R\ S \sqsubseteq (F\ S)\ Q$	The other half, with both inclusions turned around.
fusion	$\llparenthesis R \rrparenthesis\ S = \llparenthesis Q \rrparenthesis \Longleftarrow R\ S = (F\ S)\ Q$	Both halves at once: a fold followed by S collapses into a single fold, which is how an intermediate structure is got rid of.

The two Λ rows, drawn:

$$\begin{tikzcd}
F\ T \arrow[r, dashed, "F\ K"] \arrow[d, "\alpha"'] & F\ [B] \arrow[r, "F\ \ni"] \arrow[d, "\Lambda((F\ \ni)\ R)"] & F\ B \arrow[d, "R"] \\
T \arrow[r, dashed, "K"'] & [B] \arrow[r, "\ni"'] & B
\end{tikzcd}$$

The left square is that catamorphism's own defining square, $K = \llparenthesis \Lambda((F\ \ni)\ R) \rrparenthesis$, and the right one is Λ's cancellation, so the outer rectangle says $K\ \ni$ satisfies the defining equation of $\llparenthesis R \rrparenthesis$ — and uniqueness finishes it.

§14. Type functor

> F a **binary** relator: $F(R, S)$ is its action on a pair, and $F\ X$ abbreviates $F(\mathbb{1}, X)$, the F of the catamorphism section. For every object A the initial algebra is $\alpha : F(A, T\ A) \to T\ A$, among the maps. The **type functor** T acts on an arrow $R : A \to B$ by
>
> $$T\ R = \llparenthesis F(R, \mathbb{1})\ \alpha \rrparenthesis : T\ A \to T\ B$$

F is the **base functor** — one layer of the structure, acting on the recursive position — while T is the datatype itself, acting on the parameter. For cons-lists, $\mathtt{list}\ R = \llparenthesis [\mathtt{nil},\ (R \otimes \mathbb{1})\ \mathtt{cons}] \rrparenthesis$, which is $\mathtt{map}\ R$.

$$\begin{tikzcd}
F(A,\ T\ A) \arrow[r, dashed, "F(\mathbb{1},\ T\ R)"] \arrow[d, "\alpha"'] & F(A,\ T\ B) \arrow[r, "F(R,\ \mathbb{1})"] & F(B,\ T\ B) \arrow[d, "\alpha"] \\
T\ A \arrow[rr, dashed, "T\ R"'] & & T\ B
\end{tikzcd}$$

$T\ R$ is the unique arrow making it commute; there is no $\sqsubseteq$ in it.

name	law	what it says
the defining equation	$T\ R = \llparenthesis F(R, \mathbb{1})\ \alpha \rrparenthesis$	Rebuild the structure with α, applying R to the parameter on the way; that is $\mathtt{map}\ R$.
functor	$T\ \mathbb{1} = \mathbb{1}$ and $(T\ R)(T\ S) = T\ (R\ S)$	Mapping the identity changes nothing, and two maps in a row are one map.
type functor fusion	$(T\ R)\ \llparenthesis Q \rrparenthesis = \llparenthesis F(R, \mathbb{1})\ Q \rrparenthesis$	A map followed by a fold is a single fold — the mapped structure is never built.
naturality of α	$\alpha\ (T\ R) = F(R,\ T\ R)\ \alpha$	Building and then mapping is the same as mapping the parts and then building, so α is natural from $G\ R = F(R,\ T\ R)$ to T.
type relator	$(T\ R)^\circ = T\ (R^\circ)$	A datatype acts on relations, not only on maps — the map of the converse is the converse of the map.

Type functor fusion is the equality fusion row above applied to $T\ R$'s own defining algebra, whose side condition holds because F is a bifunctor — $F(R, \mathbb{1})\ F(\mathbb{1}, \llparenthesis Q \rrparenthesis) = F(R, \llparenthesis Q \rrparenthesis) = F(\mathbb{1}, \llparenthesis Q \rrparenthesis)\ F(R, \mathbb{1})$ — so it inherits that row's local-completeness requirement.

$$\begin{tikzcd}
F(A,\ T\ A) \arrow[r, "F(R,\ T\ R)"] \arrow[d, "\alpha"'] & F(B,\ T\ B) \arrow[d, "\alpha"] \\
T\ A \arrow[r, "T\ R"'] & T\ B
\end{tikzcd}$$

It commutes strictly: it is the naturality square of α.